

along with an optimised acoustic chamber enable advanced noise suppression and voice enhancement. You also get one-touch access to digital assistants like Alexa, Siri and Google. Jabra's Elite Sport range has even more awesome features including heart rate monitoring and accelerometers for tracking steps.

And, the best for the last—translation earbuds! Yes, Line Corporation and Naver Corporation have developed Mars translation earbuds that can handle real-time ear-to-ear translation in ten languages including English, Korean, Chinese and Japanese. So, the two of you who don't know each other's language can comfortably converse by using these earbuds.

Do we call these as seeables?

Like hearables that support hearing, there are several wearables that improve vision. Samsung recently demonstrated its Relúmino app and glasses, which harness the power of a smartphone to enhance viewing for those with low vision. With the comfortable and lightweight Relúmino glasses and an image processing software running on the smartphone, a person with low vision can read a book or watch TV much like anyone else does. And, because it uses a normal smartphone, the solution is pretty inexpensive too.

The app and glasses can be used in four modes: Regular mode to make shapes and outlines more prominent, Color Invert mode to read text, Partial Vision mode for those with central or peripheral vision loss, and Display Color Filter mode that reduces glare and works like sunglasses when outdoors.

Eyewear is not just necessity-driven—there are lots of fun-packed options too. For instance, Spectacles by Snap is a snazzy pair of sunglasses with two tiny cameras fitted on either side. It helps you record events at the press of a button, so you can relive them from your perspective at any time in the future. To set privacy fears at ease, the spectacles light up to let others know that you are recording.

There are also innumerable glasses for those who love the virtual world. Examples include HTC Vive, Oculus



Spectacles that record video (Courtesy: Snap Inc.)

Rift, Vuzix smart glasses, Samsung's Gear VR, Google Cardboard and Black-box VR. However, virtual reality is a separate subject in its own right!

Coaching and training, for the love of sports

A slew of wearables have changed the way people train and play. Right from gait monitoring to game analysis, wearables are helping players to understand and improve their game further.

For athletes, there are tools like Run Free Pro, a sensor-studded headphone that measures speed, distance, cadence, step length and width, vertical oscillation, head-tilt angle, balance, consistency and other factors. Working with a smartphone app, the headphone provides the runner with real-time coaching! The headphones are comfortable to wear and sweat-proof too.

Sensoria and Vivobarefoot have teamed up to develop smart shoes with sensors threaded through their sole. So, apart from speed, pace and cadence, the runner also gets data on foot landing spot, ground time and impact score. There are a lot more wearables for runners from companies like Moov and Milestone, and likewise, for every other sport including football and swimming.

There are wearables that improve players' safety too. Prevent Biometrics hopes to prevent football players' common problem of concussions using its Head Impact Monitor System (HIMS), a concussion-sensing mouth-guard. When a player is hit hard, three-channel accelerometers of the device record the impact. The software differentiates between light impacts and serious ones. An alert is issued if the impact is above a certain acceptable level. The mouth-guard lights up and a notification is sent to the associ-



Sensors on the sole collect data on foot landing, ground time, impact, etc (Source: Vivobarefoot)

ated device so that the coach or other emergency personnel know that the player needs to stop playing for safety reasons.

Stay safe, healthy and beautiful

Improving personal safety. ZTE, Wearsafe and Qualcomm have teamed up to create unobtrusive wearable devices that improve personal safety by instantly connecting the wearer with first responders, family and other emergency contacts at the push of a button. Qualcomm's mobile processor Snapdragon Wear 1100 will ensure that these devices are low-power and have next-gen LTE support. While ZTE will provide the rest of the hardware, Wearsafe will bring in the personal security platform, which specialises in giving the emergency contact relevant information about the wearer, such as his GPS location, a 60-second buffered audio clipping of what happened before the button was pressed, and a live audio stream of current happenings.